WORKSHEET

Chapter 2: Trigonometric identities using De Moivre's theorem

1 **a** Expand $[\cos(\theta) + i\sin(\theta)]^3$ using the binomial theorem.

b Show that $\cos^3(\theta) = \cos(3\theta) + 3\cos(\theta)$.

c Show that $\sin(3\theta) = 3 - 4\sin^3(\theta)$.

2 **a** Expand $[\cos(\theta) + i\sin(\theta)]^2$ in two ways.

b Hence, show that

i $\cos^2(\theta) - \sin^2(\theta) = \cos(2\theta)$

ii $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$

3 Use the expansion of $[\cos(\theta) + i\sin(\theta)]^3$ to show that $\cos(3\theta) = \cos^3(\theta) - 3\cos(\theta)\sin^2(\theta)$.

© Cengage Learning Australia Pty Ltd 2014

4 Use the expansion of $[\cos(\theta) + i\sin(\theta)]^4$ to show that $\cos(4\theta) = \cos^4(\theta) - 6\cos^2(\theta)\sin^2(\theta) + \sin^4(\theta)$.

5 Use the expansion of $[\cos(\theta) + i\sin(\theta)]^4$ to show that $\sin(4\theta) = 4\cos^3(\theta)\sin(\theta) - 4\cos(\theta)\sin^3(\theta)$.

6 Use expansions of $[\cos(\theta) + i\sin(\theta)]^5$ to prove that:

a $\cos(5\theta) = 16\cos^5(\theta) - 20\cos^3(\theta) + 5\cos(\theta)$

b $\sin(5\theta) = 16\sin^5(\theta) - 20\sin^3(\theta) + 5\sin(\theta)$

7 **a** Complete the statement $\tan(5\theta) = \dfrac{\ldots(5\theta)}{\cos(\ldots)}$.

b Show that $\cos(5\theta) = \cos^5(\theta) - 10\cos^3(\theta)\sin^2(\theta) + 5\cos(\theta)\sin^4(\theta)$.

© Cengage Learning Australia Pty Ltd 2014

c Show that $\sin(5\theta) = \sin^5(\theta) - 10\cos^2(\theta)\sin^3(\theta) + 5\cos^4(\theta)\sin(\theta)$.

d Hence, prove that $\tan(5\theta) = \dfrac{5\tan(\theta) - 10\tan^3(\theta) + \tan^5(\theta)}{5\tan^4(\theta) - 10\tan^2(\theta) + 1}$.

8 a Express $\sin(60°)$ in terms of $\sin(20°)$.

b Given that $\sin(20°) \approx 0.342\,020\,1$, find a value for $\sin(60°)$ without using a calculator or tables.

9 a Use the expansion of $[\cos(\theta) + i\sin(\theta)]^3$ to show that $4\cos^3(\theta) = \cos(3\theta) + 3\cos(\theta)$.

b Hence, find the exact value of $\int_0^{\frac{\pi}{3}} 4\sin^3(\theta)\,d\theta$.

Answers

1 **a** $[\cos(\theta) + i\sin(\theta)]^3 = [\cos^3(\theta) - 3\cos(\theta)\sin^2(\theta)] + i[3\cos^2(\theta)\sin(\theta) - \sin^3(\theta)]$

b Demonstration

c Demonstration

2 **a** $[\cos(\theta) + i\sin(\theta)]^2 = \cos^2(\theta) - \sin^2(\theta) + 2i\cos(\theta)\sin(\theta)$

$[\cos(\theta) + i\sin(\theta)]^2 = \cos 2\theta + i\sin 2\theta$

b **i** Equate real parts of equations in part **a**.

ii Equate imaginary parts of equations in part **a**.

3 Equate real parts.

4 Equate real parts.

5 Equate imaginary parts.

6 **a** Proof

b Proof

7 **a** $\tan(5\theta) = \dfrac{\sin(5\theta)}{\cos(5\theta)}$

b Demonstration

c Demonstration

d Proof

8 **a** $\sin(60°) = 3\sin(20°) - 4\sin^3(20°)$

b 0.866 025 4

9 **a** Expand $[\cos(\theta) + i\sin(\theta)]^3$ and rearrange the result.

b $2\frac{1}{2}$

© Cengage Learning Australia Pty Ltd 2014